

Pharmacology: Multiple Sclerosis (MS)

Maria A. Pino, Ph.D
Department of Clinical Specialties
mpino@nyit.edu

Session Objectives

- 1. Review the pathogenesis of Multiple Sclerosis (MS).**
- 2. Describe the drugs used for acute flare-ups of MS, mainly glucocorticoids. Identify adverse effects.**
- 3. Describe the use of immunotherapy for the treatment of MS. Focus on the pharmacological target and adverse effects.**
- 4. Describe supplemental treatments for MS. Include mechanisms and adverse effects.**

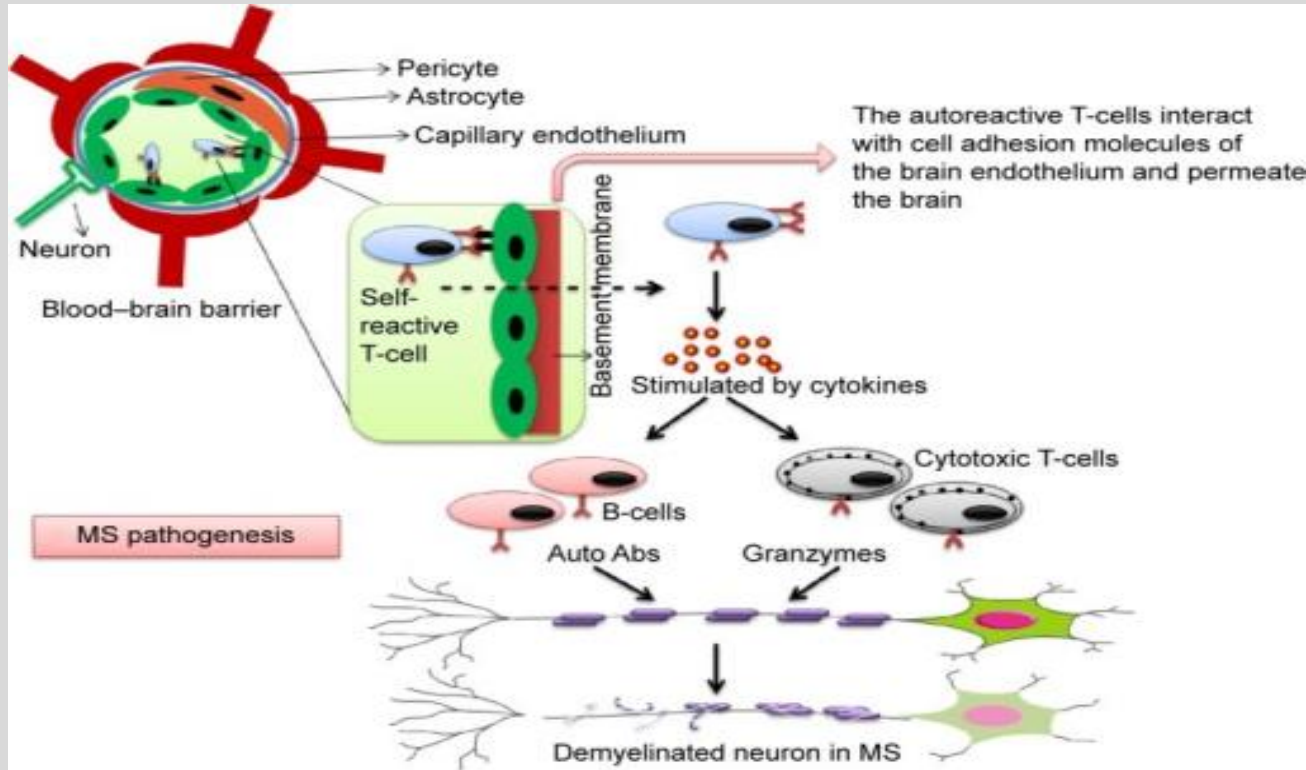
Multiple Sclerosis (MS)

- Relapsing or progressive autoimmune disease
- Involves inflammation, demyelination, plaques, neuronal loss.
- B cell and **T cell mediated (TH1), (TH17)**.
- Sensory symptoms, optic neuritis, weakness, spasticity, fatigue, urinary incontinence, depression, erectile dysfunction
- Female > males
- Young > older adults
- Genetics, smoking, virus exposure
- Low vitamin D levels?

Pathogenesis of Multiple Sclerosis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Multiple Sclerosis (MS) Drugs

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Acute: glucocorticoids (prednisone, **IV methylprednisolone**). Optic neuritis treatment.
- Relapsing-remitting: Immunotherapy. **Slow disease progression.**
 - Anti-CD52 drug (IV)
 - Dimethyl fumarate (oral)
 - Fingolimod (oral)
 - Glatiramer acetate (sq)
 - Interferon beta (sq, IM)
 - Natalizumab (IV)
 - Anti-CD20 drugs (IV)
 - Teriflunomide (oral)
 - Mitoxantrone (antineoplastic, secondary-progressive MS)
 - Cladribine (oral)

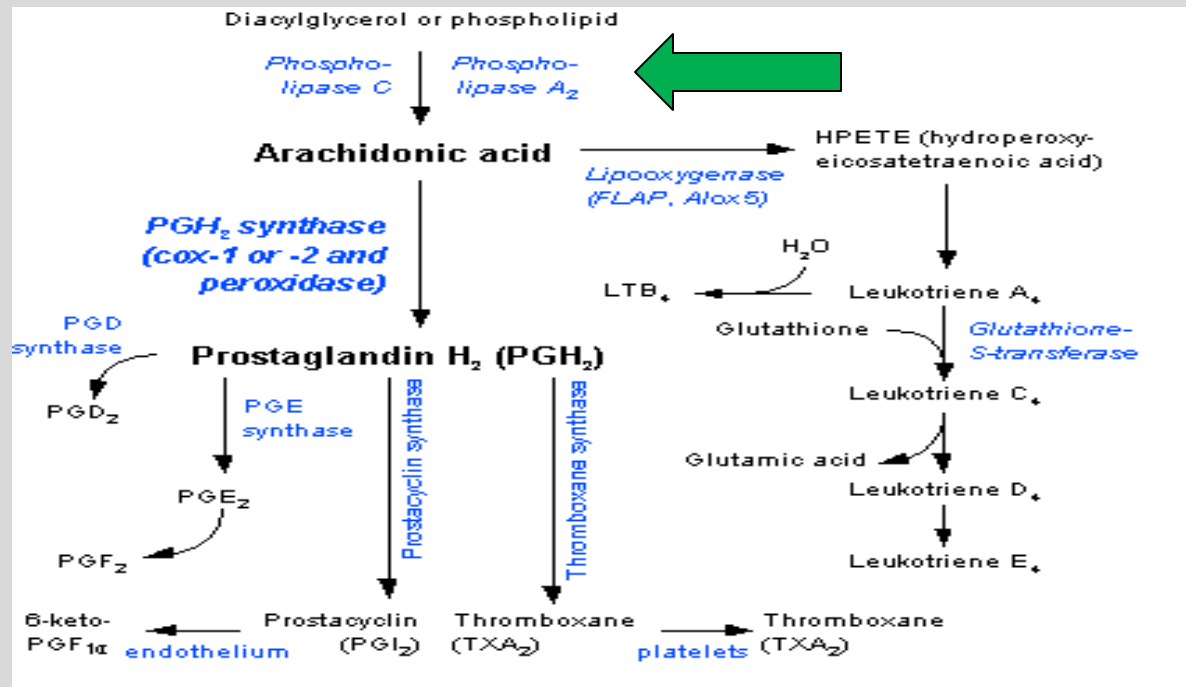
Glucocorticoids

- Stimulate synthesis of lipocortin

- Lipocortin inhibits Phospholipase A₂
 - Blocks starting point for prostaglandin, leukotriene synthesis

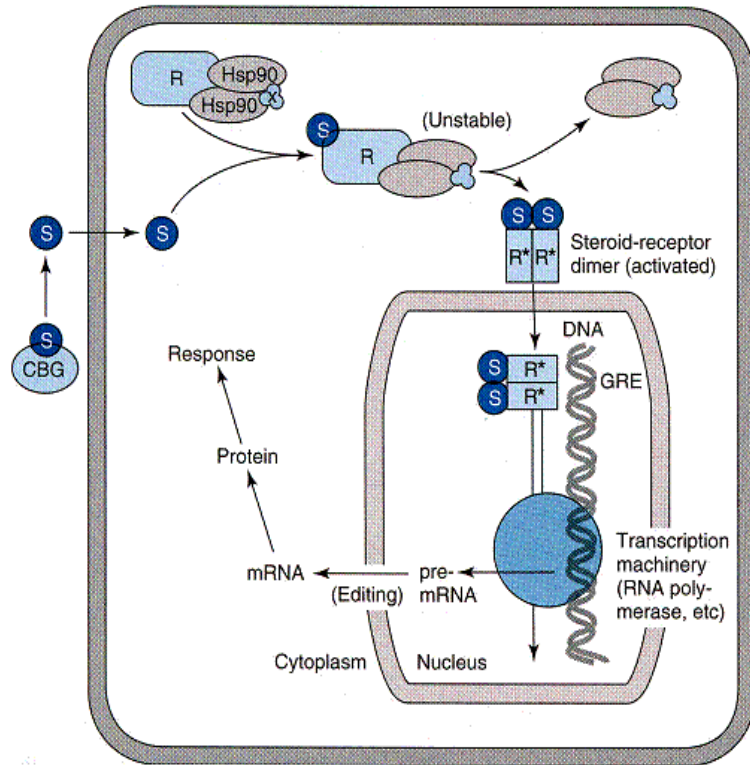
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



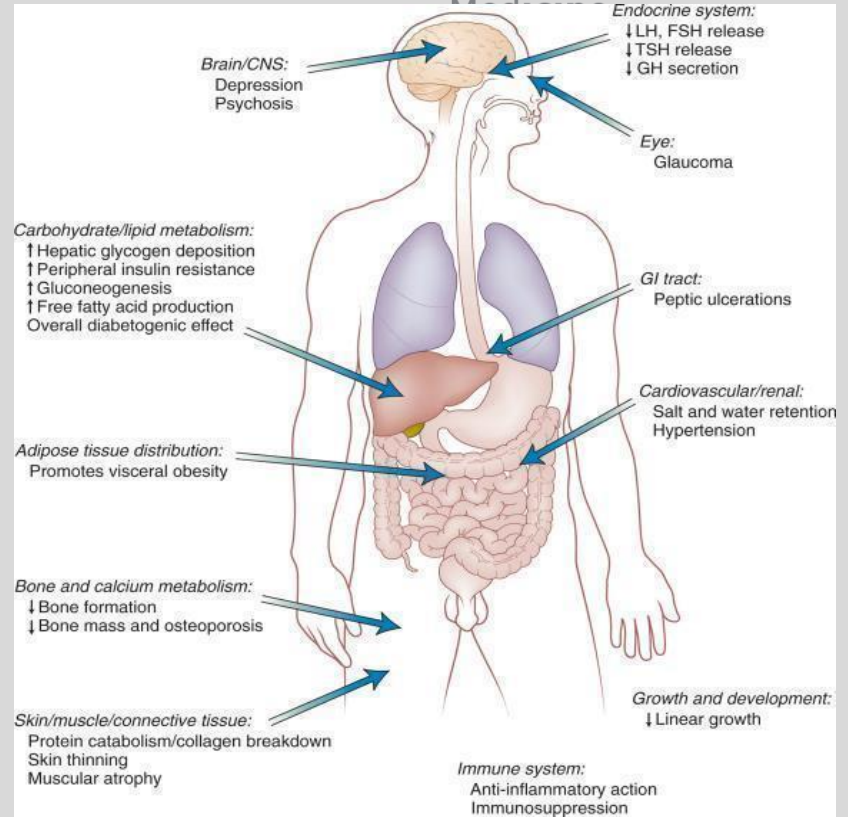
Glucocorticoids

HPA suppression: taper dose slowly



NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic



Alemtuzumab: Anti-CD52

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

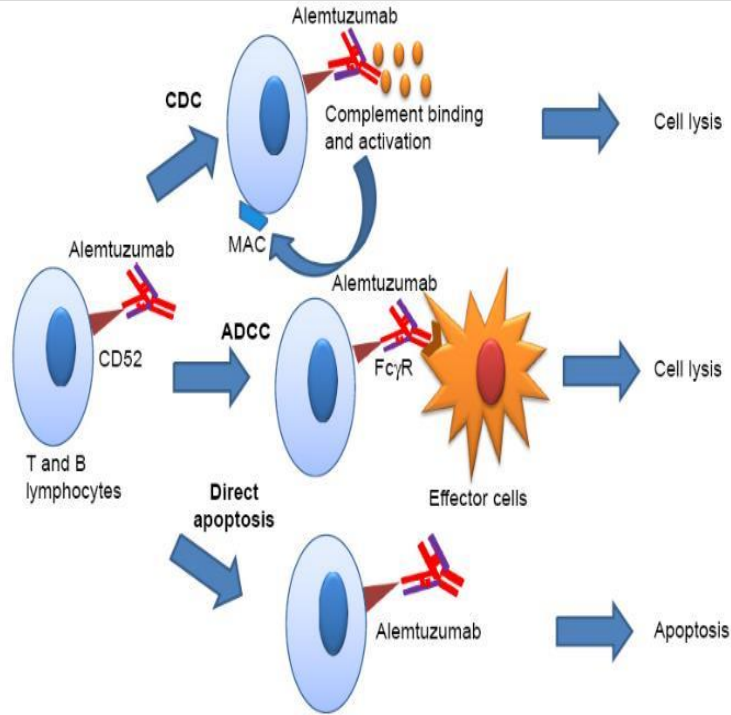


Figure 1 Alemtuzumab-mediated cytotoxicity and apoptosis of T- and B-lymphocytes.

Abbreviations: ADCC, antibody-dependent cell-mediated cytotoxicity; CDC, complement-dependent cytotoxicity; MAC, membrane attack complex; FcγR, Fc-gamma receptor.

AE:

- Pregnancy warning
- Depletes lymphocytes
- Autoimmune reactions
- Infusion reactions

Other uses:

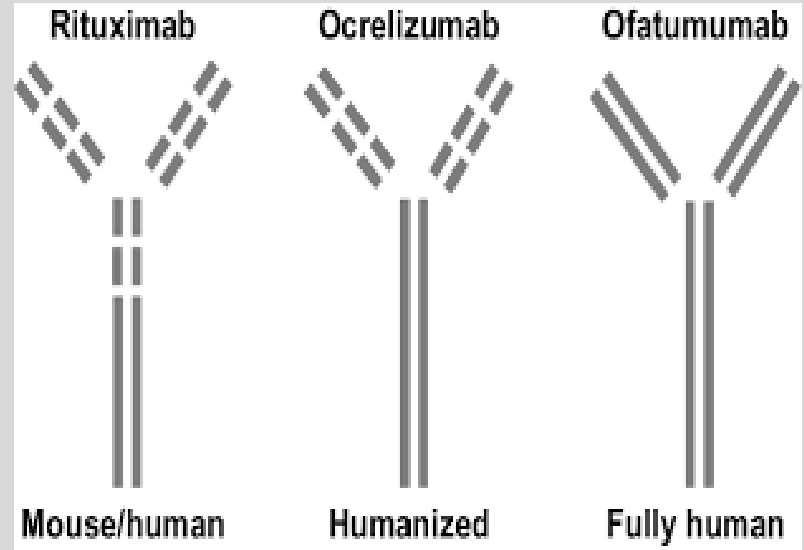
- Chronic lymphocytic leukemia (CLL)
- T-cell lymphoma.

Monoclonal Antibodies

- -umab = human (only human protein)
- -momab = mouse (murine)
- -ximab = chimeric (mix of mouse and human)
- -zumab = humanized (mix of mouse and human, less mouse protein compared to chimeric)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

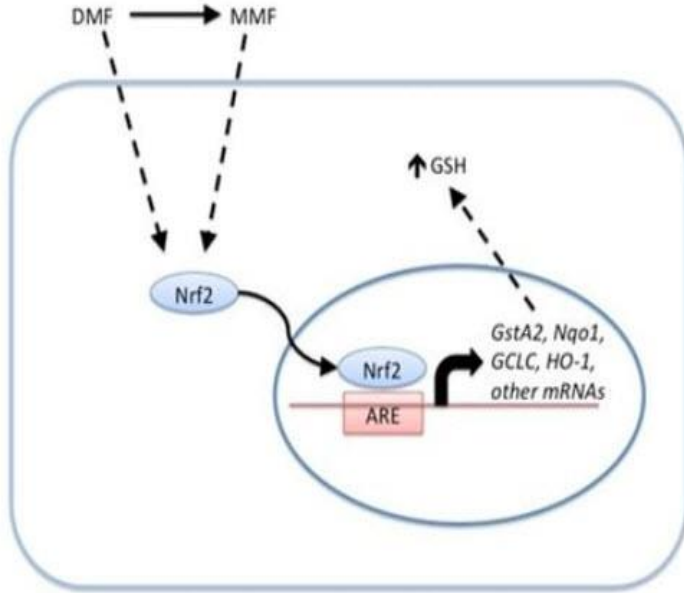
College of Osteopathic
Medicine



Dimethyl fumarate

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

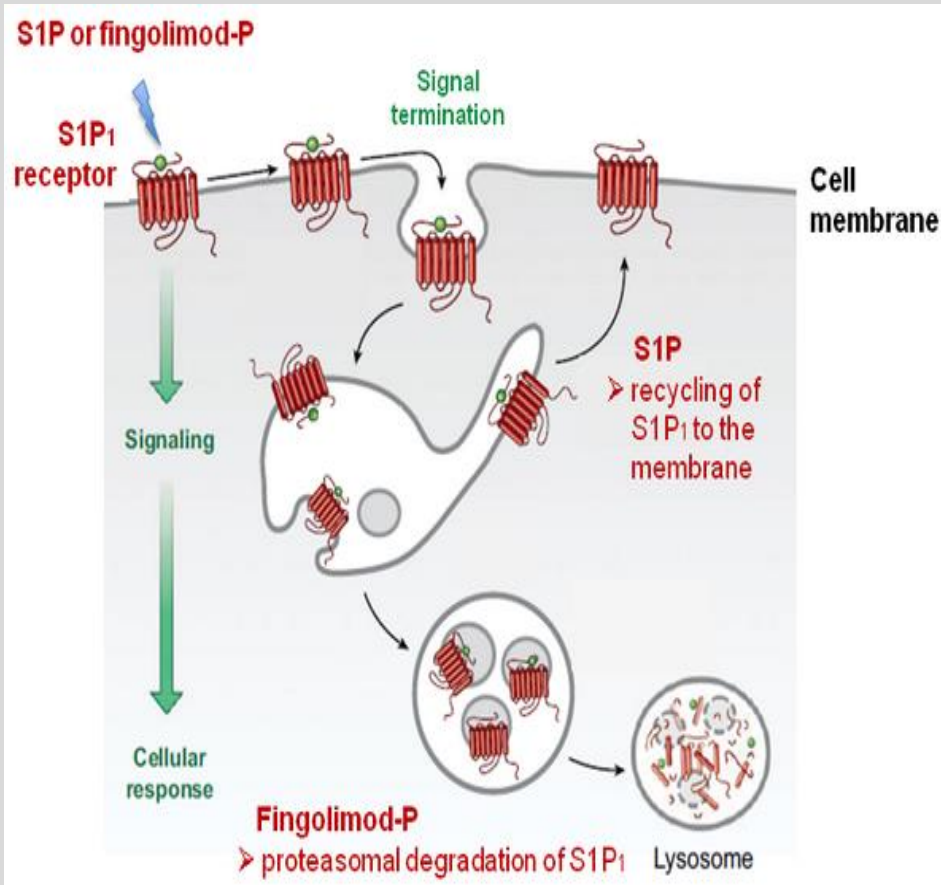


- Nrf2 activator via metabolite monomethyl fumarate.
- AE: GI upset, flushing, Progressive multifocal leukoencephalopathy (PML), elevated liver enzymes, increased infection risk.

Fingolimod

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



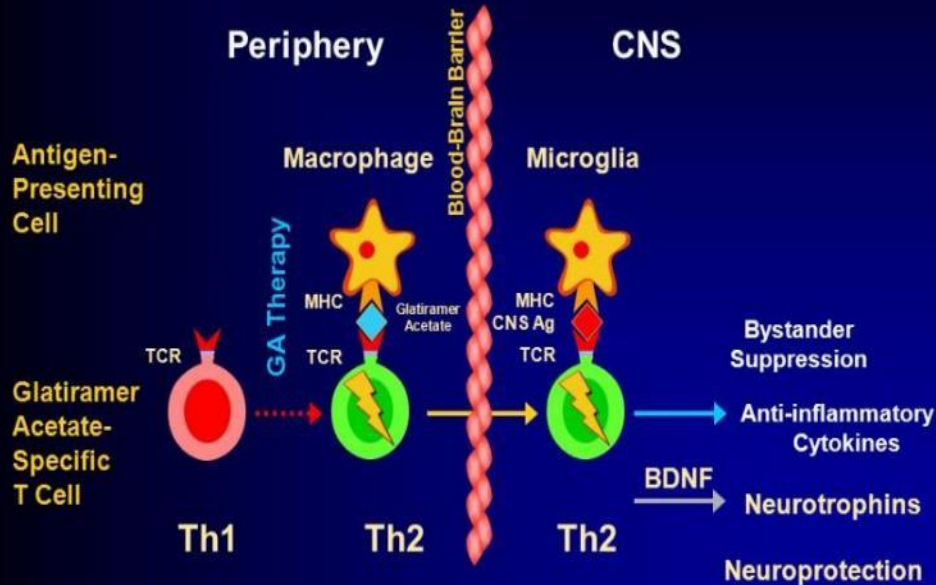
- Sequestering of lymphocytes
- AE: Headache, **bradycardia**, activation of infections, PML.
- Basal cell carcinoma
- Test VZV antibody
- Evaluate EKG (at first dose, re-initiation).

Glatiramer acetate

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Mode of Action of Glatiramer Acetate: Bystander Suppression and Neuroprotection



Adapted with permission from Neuhaus O et al. *Neurology*. 2001;56:702-708.
Ziemssen T et al. *Brain*. 2002;125:2381-2391.

- **AE: Injection site reactions**
- **Post-injection reaction: flush, SOB, palpitations ~occurs once.**
- **Lipoatrophy (SQ injection effect)**

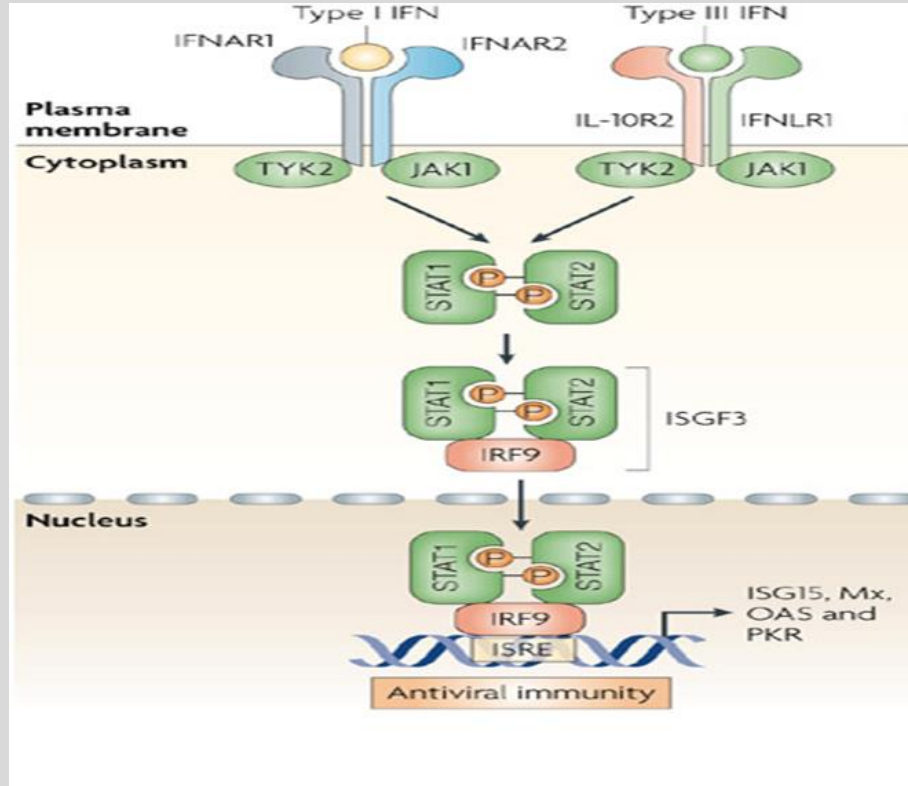
Why infusion reactions?

How to manage infusion reactions?

Interferon B1a

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

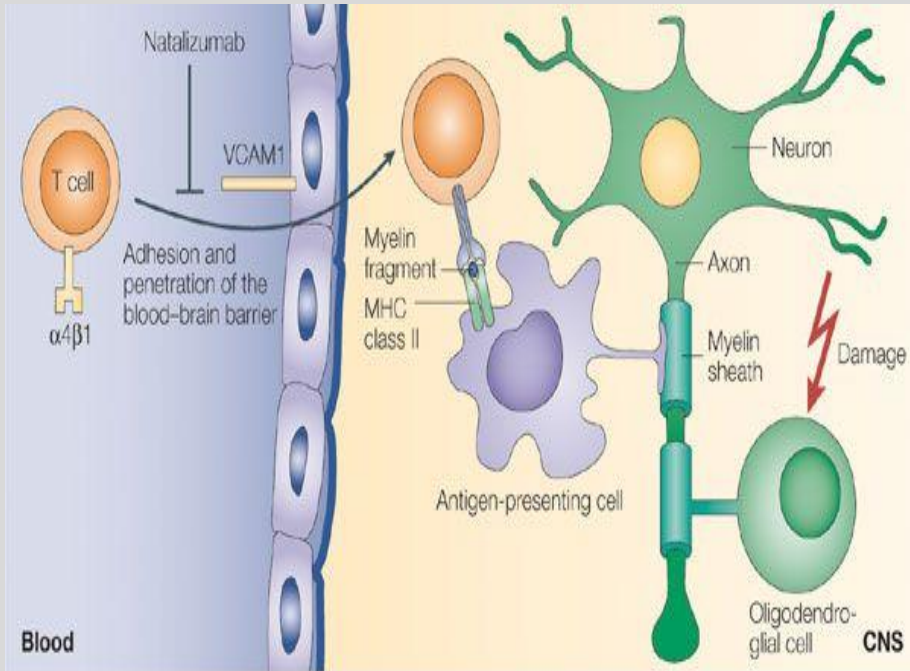


- Cytokine, Via JAK-STAT pathway
- MS: inhibits T-cell activation and migration across BBB.
- AE: flu-like symptoms, depression, injection reactions, anemia.
- Also treats hepatitis B, C.

Natalizumab

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



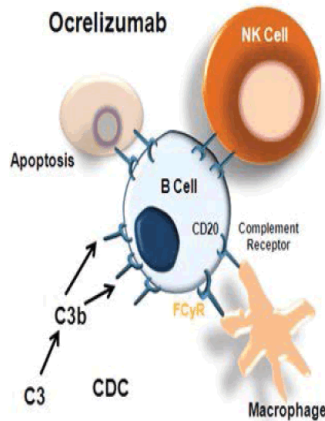
Nature Reviews | Drug Discovery

- **Binds ($\alpha 4$, $\beta 1$) receptor, stops T-lymphocyte adhesion to VCAM**
- **AE: infusion reactions, PML, hepatotoxicity, infection risk**
- **Also used for Crohn's disease**

Anti-CD20 drugs

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Antibody	Type	ADCC	CDC
Rituximab	Chimeric IgG1	++	++
Ocrelizumab	Fully humanized IgG1	+++	+/-

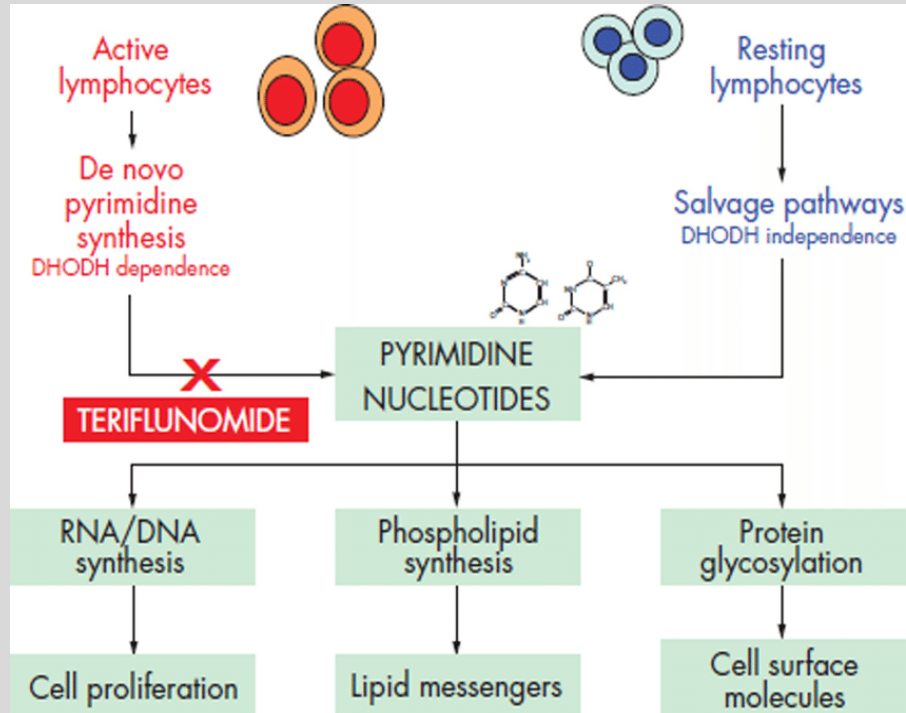
Figure 5: Ocrelizumab vs. Rituximab characteristics [17,18].

- **AE: infusion reactions, respiratory infections, activation of viral infections, such as hepatitis B**
- **Rituximab (B-cell lymphoma treatment)**
- **Ocrelizumab: Humanized**
- **Ofatumumab: Fully human**
- **Ublituximab: twice yearly infusion**

Teriflunomide

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

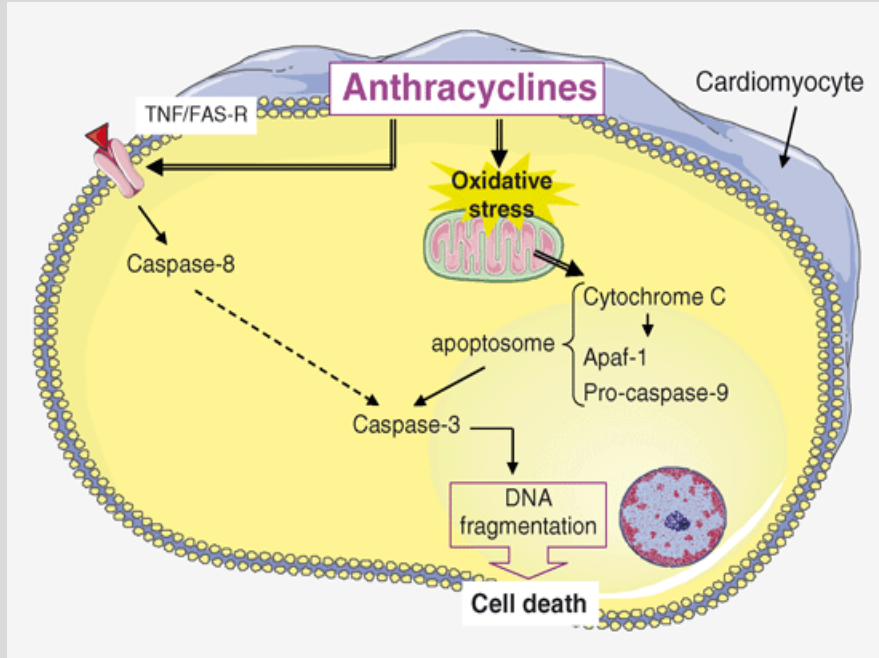


- **Metabolite of leflunomide (used in RA)**
- **AE: infections, hepatotoxicity, and teratogenicity**
- **(Continue contraception 2 years post-therapy)**

Mitoxantrone: Secondary Progressive MS

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



- Intercalating agent
- “Doxo**rub**icin” like. Anthracycline
- DNA strand scission via effects on topoisomerase II
- Binding to membranes alters fluidity
- Generation of oxygen radicals
- AE: cardio, N/V, alopecia, immunosuppression
- Also treats acute myeloid leukemia (AML), prostate, breast cancer

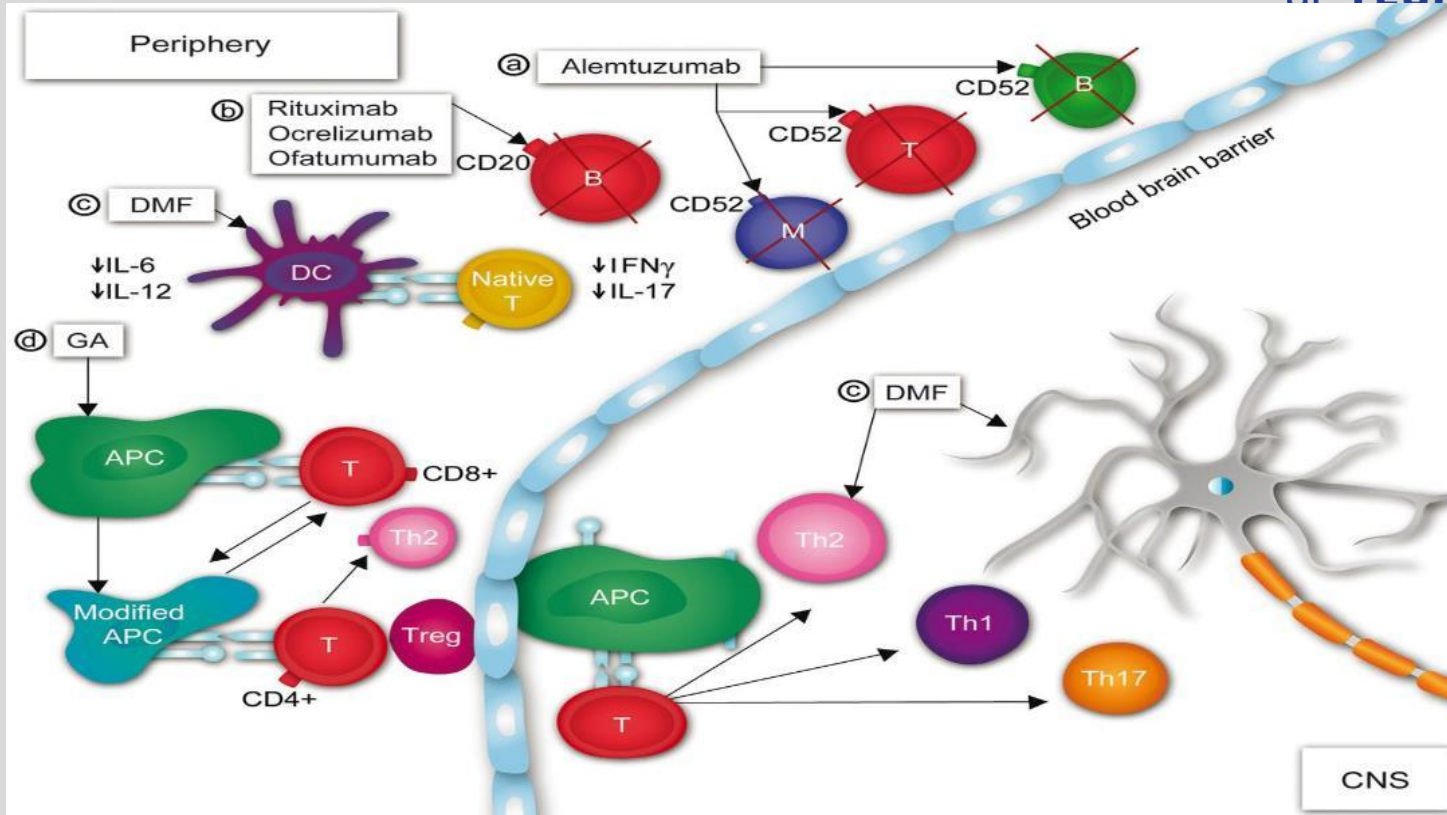
Cladribine

- **Synthetic deoxyadenosine (purine) analog.
Antipyrimidine**
- **Antimetabolite chemotherapeutic (treats hairy
cell leukemia)**
- **Greater effect on B cells vs T cells; allows for
apoptosis of lymphocytes.**
- **AE: Lymphopenia and infections (VZV)**

Summary: MS agents

NEW YORK INSTITUTE
OF TECHNOLOGY

Osteopathic



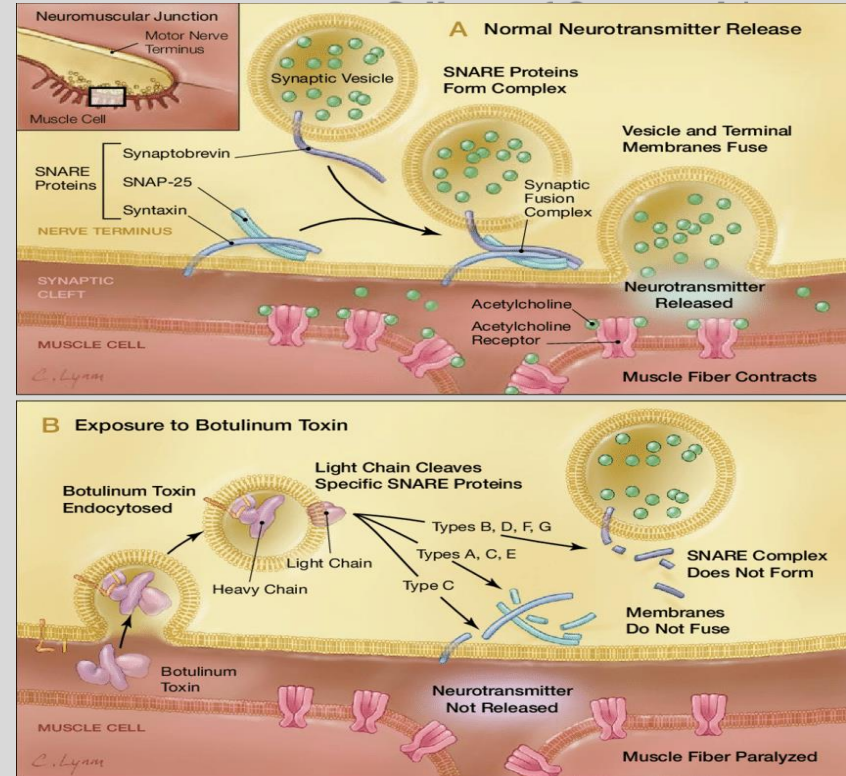
Incontinence

- M₃ blockers: oxybutynin, tolterodine, darifenacin
- Gq mechanism. AE??
- Botulinum toxin: cleaves SNARE protein (SNAP-25) responsible for vesicle fusion.

AE: Paralysis

Other uses: strabismus, blepharospasm, wrinkles, migraines, cervical dystonia

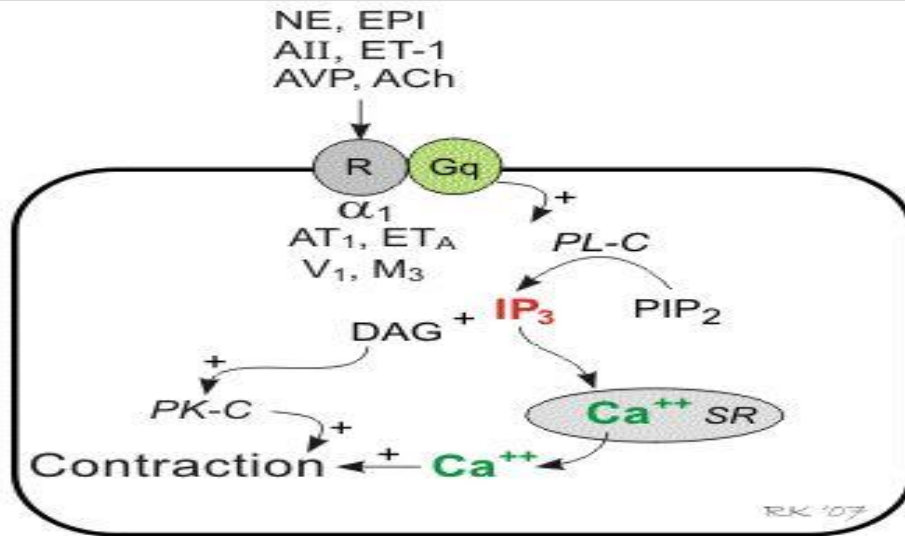
NEW YORK INSTITUTE
OF TECHNOLOGY



G-receptor proteins (Gq)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



R, receptor; Gq, phospholipase C-coupled Gq-protein; PL-C, phospholipase C; PIP₂, phosphatidylinositol biphosphate; IP₃, inositol triphosphate; DAG, diacylglycerol; PK-C, protein kinase C; SR, sarcoplasmic reticulum; NE, norepinephrine; AII, angiotensin II; ET-1, endothelin-1; AVP, arginine vasopressin; ACh, acetylcholine; α_1 , α_1 -adrenoceptor; AT_1 , type 1 angiotensin receptor; ET_A , type A endothelin receptor; V_1 , vasopressin type 1 receptor; M_3 , muscarinic type 3 receptor.

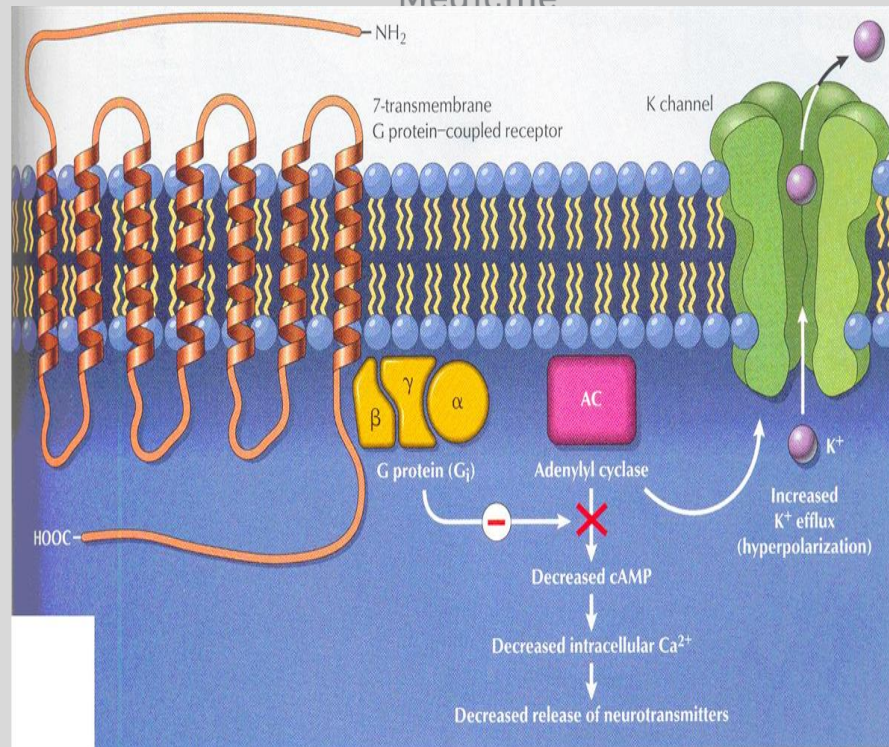
Spasticity

Goal: Reduce muscle tone

- Tizanidine (agonist of α_2 adrenergic receptor). Like clonidine
- Baclofen (GABA_B agonist): G protein (Gi) mechanism
- AE: CNS depression

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Depression and Fatigue

**NEW YORK INSTITUTE
OF TECHNOLOGY**

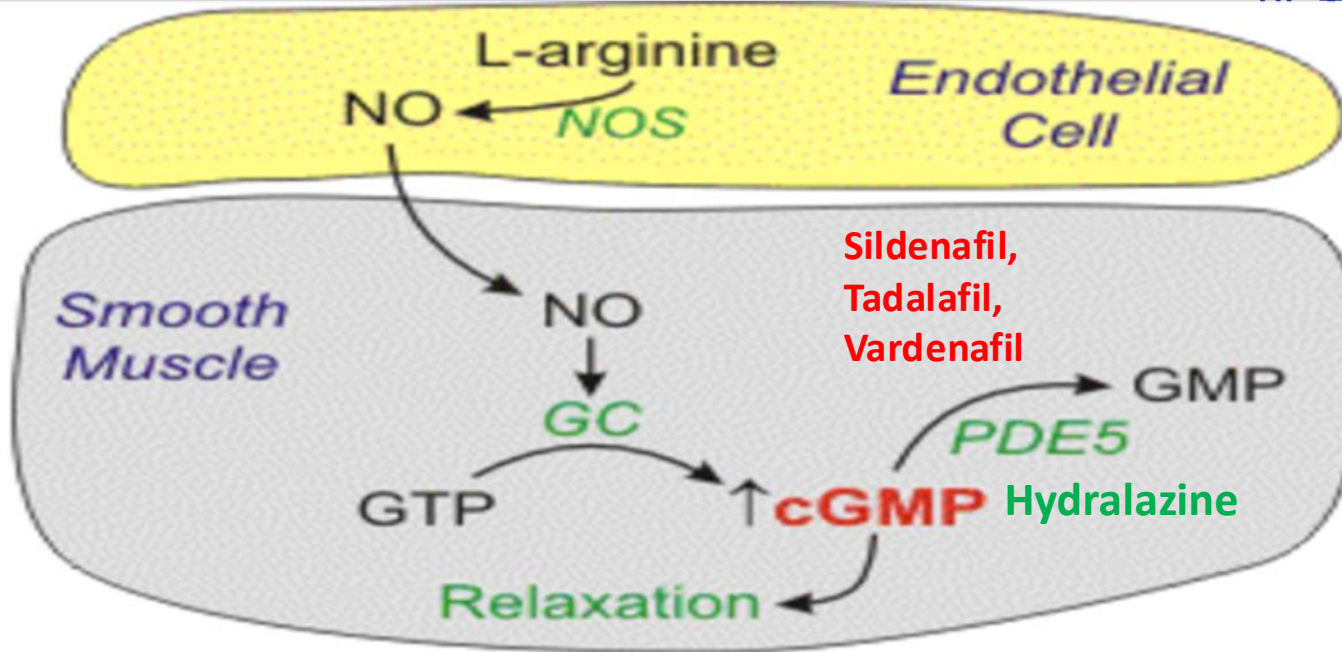
College of Osteopathic
Medicine

- **SSRIs (sertraline, fluoxetine, citalopram)**
- **SNRIs (duloxetine)**
- **Bupropion (also nicotinic antagonist; smoking cessation)**
- **Tricyclic Antidepressants (also helps urinary incontinence) M₃ blockade. Doxepin, amitriptyline, imipramine**
- **Amphetamines (methylphenidate, modafinil)**

Erectile Dysfunction

NEW YORK INSTITUTE
OF TECHNOLOGY

Department of Osteopathic
Medicine



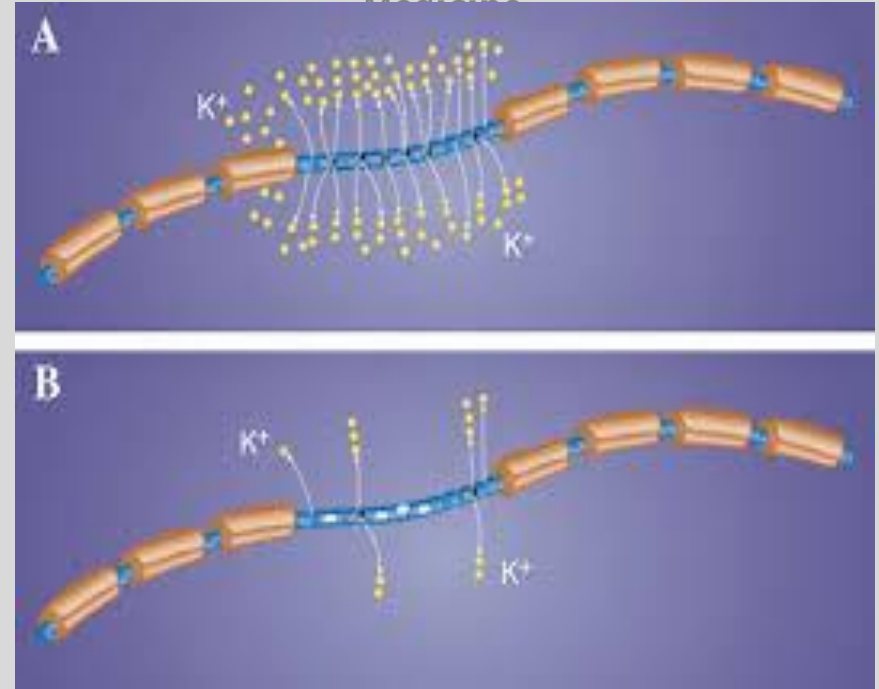
Abbreviations: NO, nitric oxide; NOS, nitric oxide synthase; GC, guanylyl cyclase; PDE5, cGMP-dependent phosphodiesterase (type 5)

Other Drugs

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- **Dalfampridine: 4-aminopyridine. Block potassium channels. Increases action potential duration. Improved walking.**



Summary Slide

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Describe the drugs used for acute and relapsing MS. Focus on the pharmacological target and adverse effects.
- Describe supplemental treatments for MS. Include mechanisms and adverse effects

Core References:

1. Brunton and Knollman; Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 39: Immunotherapy for Multiple Sclerosis
2. Katzung and Vanderah; Basic & Clinical Pharmacology, 16e, 2024; Chapter 55: Immunopharmacology
3. Multiple Sclerosis drug summary and practice questions (Maria A. Pino, Ph.D)

Lecture Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>